

Site-specific irrigation scheduling using one-layer soil hydraulic properties and inverse modeling

Hemendra Kumar^{a,*}, Puneet Srivastava^{b,c}, Jasmeet Lamba^a, Efstathios Diamantopoulos^d, Brenda Ortiz^e, Guilherme Morata^f, Bijoychandra Takhellambam^a, Luca Bondesan^e

^a Department of Biosystems Engineering, Auburn University, Auburn, Alabama 36849, USA

^b College of Agriculture and Natural Resources, University of Maryland, College Park, Maryland 20742, USA

^c Maryland Agricultural Experiment Station, University of Maryland, College Park, Maryland 20742, USA

^d Chair of Soil Physics, University of Bayreuth, Bayreuth, Germany

^e Department of Crop Soil and Environmental Science, Auburn University, Auburn, Alabama 36849, USA

^f Alabama Agricultural Experiment Station, Auburn University, Auburn, Alabama 36849, USA

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ABSTRACT

Site-specific soil hydraulic properties (SHPs) are critical for determining irrigation thresholds and adopting the best irrigation practices. In this study, we optimized two single-layer SHPs for two management zones (namely, zone 1 and zone 2) delineated in a crop field using topographic attributes, soil texture, and historical crop yield to demonstrate the need for zone-specific SHPs. The inverse HYDRUS-1D model was used to optimize a single-layer zone-specific SHPs using multilayered observed soil water pressure heads (h) measured in both zones. Statistical indices for goodness of fit showed that a single layer SHPs simulate h during the growing season well. The irrigation thresholds and amounts determined using zone-specific optimized SHPs were used to trigger irrigations. Different irrigation scenarios using 15 cm, 30 cm, and 60 cm soil depths were developed using the zone-specific optimized SHPs in the HYDRUS-1D model to meet irrigation water demand and reduce water stress. The ratio of actual root water uptake (ARWU) and potential root water uptake (PRWU) were calculated to compare improvements in satisfying daily water demand and to reduce the water stress under different irrigation scenarios. Out of the total irrigation water triggered during growing seasons, greater than 61 % and 75 % in zone 1 was triggered in July 2018 and in May-June 2019, respectively. However, greater than 73 % in July 2018 and greater than 72 % in May-June 2019 were triggered in zone 2. Based on the simulations of various irrigation scenarios in HYDRUS-1D, we found that installing sensors at 15 cm or 30 cm can help meet the daily irrigation water demand on time than sensors at 60 cm depth. This study discussed the importance of critical water demands due to the distribution of precipitation over the growing season and need of zone-specific optimized SHPs for site-specific irrigation management within the field.

Abbreviations: ARWU, Actual Root Water Uptake; BC, Boundary Condition; IA, Index of Agreement; LAI, Leaf Area Index; MESA, Mid-Elevation Spray Application; MZA, Management Zone Analyst; NI, No Irrigation; PRWU, Potential Root Water Uptake; RMSE, Root Mean Square Error; RRMSE, Relative Root Mean Square Error; RWU, Root Water Uptake; SCE-UA, Shuffled Complex Evolution; SWD, Soil Water Depletion; TWI, Topographic Wetness Index; HCC, Hydraulic Conductivity Curve; SHP, Soil Hydraulic Property; SWRC, Soil Water Retention Curve; VRI, Variable Rate Irrigation; K , Unsaturated hydraulic conductivity; K_s , Saturated hydraulic conductivity; θ , Soil moisture/soil water content; h , Soil water pressure head; S_e , relative saturation or effective water content; θ_s , Saturated water content; θ_r , Residual water content; α, n, m, l , Shape parameters in van Genuchten model for SHP; θ_{SWD} , Soil water content at SWD; h_{SWD} , Soil water pressure at SWD; θ_{FC} , Soil water content at field capacity; θ_{pwp} , Soil water content at permanent wilting point; I_{SWD} , Irrigation depth or amount at SWD; S , Sink term in water flow equation; z , Vertical spatial coordinate; Z , Thickness of soil layer; $\alpha(h)$, Root water uptake stress response function; $\beta(z)$, Normalized water uptake distribution in sink term; T_p , Potential transpiration rate in sink term; pF , Logarithmic function of absolute h in cm; j , Number of observation point; nn , Total number of measurements at each soil depth; x , Vector of soil hydraulic parameters in van Genuchten model; $\bar{pF}_j$, Vector of daily observation of h as pF ; $\bar{pF}_j$, Vector of daily HYDRUS-1D predictions of h as pF ; 0Lag, No delay in triggering irrigation; 1Lag, One-day delay in triggering irrigation; 2Lag, Two-day delay in triggering irrigation; 3Lag, Three-day delay in triggering irrigation; 4Lag, Four-day delay in triggering irrigation; 5Lag, Five-day delay in triggering irrigation.

* Correspondence to: Auburn University, 200 Corley Building, Auburn, Alabama 36849, USA.

E-mail address: hzk0065@auburn.edu (H. Kumar).

¹ ORCID: 0000-0002-0469-0569.

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1. Introduction

Population growth and climate change are expected to increase freshwater demands (FAO, 2016; Maja and Ayano, 2021). Furthermore, at the same time, more than 40 % of the earth's population is already experiencing water scarcity problems (Kummu et al., 2016). Approximately 70 % of available freshwater withdrawal is for meeting irrigation needs in agricultural production to meet the food demands. Thus, any improvements in the current irrigation practices can help conserve limited water resources and promote sustainable agriculture (Himanshu et al., 2021; FAO, 2016). Depending on climate conditions, irrigation water demands vary greatly from arid to humid climate regions (O'Shaughnessy et al., 2018; Rolle et al., 2021). In humid climate regions, irrigation is needed due to non-uniform rainfall distribution (Sangha et al., 2020), distribution of precipitation over the growing season (Filho et al., 2020), and soil moisture variation within a crop field during a growing season (Kumar et al., 2021b). Within-field soil moisture variation can be due to the variability in soil properties, vegetation, and topographic gradients (Fontanet et al., 2020; Kumar et al., 2021b), which affects the irrigation rate, time, and depth during the crop growing period. Since uniform irrigation does not account for within-field variability in soil moisture and applies fixed irrigation amounts over the field, it can result in over and under application of water in different parts of a field. The water is expensive and limited, irrigation needs to be optimized for applying the right amount at the right place and at the right time. Because of this, although, currently uniform irrigation is the most common practice (Kumar et al., 2021b), farmers are increasingly adopting modern irrigation practices (Lena et al., 2020), for example, variable rate irrigation (VRI) based on management zones within a crop field.

To adopt VRI as the best irrigation practice, different management zones can be delineated within a field based on variations in crop yield, soil type, and topography (Kumar et al., 2021a), which has the potential to conserve water by meeting zone-specific crop water demands (Liakos et al., 2017; Lowrance et al., 2016). A VRI technology can trigger irrigations at zone-specific irrigation thresholds depending on plant available water within specific zones, which helps to increase irrigation water use efficiency, reduce leaching and runoff, reduce water stress, maximize crop yield and profitability, and reduce pressure on freshwater resources (Evans and Sadler, 2008; Stone et al., 2019; Yost et al., 2019). Sui and Yan (2017) found increased yield using with VRI than uniform irrigation in a humid climate. However, precise irrigation to meet zone-specific crop water demand requires continuous monitoring of soil water dynamics within the zones (Fontanet et al., 2020; Reyes et al., 2019).

In the past ten years, sensor-based irrigation method has received a lot of attention since this approach provides real-time soil water status (as soil water content- θ or soil water pressure- h), which can be used for uniform irrigation and VRI. Sensor locations are selected based on variabilities (soil, topography, yield, etc.) in the field. Detailed information about sensor placement can be found in Irmak et al. (2012). Monitoring soil water dynamics in real-time not only improves irrigation timing and depth (Irmak et al., 2010), it also permits adequate water distribution using the VRI technique (O'Shaughnessy et al., 2018; Sharma and Irmak, 2020). To reduce crop water stress and potential crop yield losses, predefined soil water depletion (SWD) methods have been used by irrigation experts to determine irrigation thresholds using h or θ (Gu et al., 2020), where irrigation threshold, amount, and time are determined by maximum allowable SWD in each management zone (Allen et al., 1998; Irmak, 2016; Sui and Yan, 2017).

In sensor-based irrigation, soil water dynamics in the soil profiles can be explained using knowledge on the soil hydraulic properties (SHPs), i. e., the soil water retention curve (SWRC) and the hydraulic conductivity curve (HCC). Soil hydraulic properties are required to understand root water uptake (Simunek and Hopmans, 2009) and to determine irrigation thresholds (Gu et al., 2020; Wang et al., 2017). Assuming one SWRC and

HCC throughout the field does not account for zone-specific soil water dynamics and zone-specific characteristics in a crop field. Unfortunately, determination of both SWRC and HCC and use of them to determine irrigation thresholds is rarely applied in irrigation applications. Even when these relationships are applied for determining irrigation thresholds, investigations that optimize zone-specific SHPs (SWRC and HCC) and quantify their effect on irrigation threshold determination have not been conducted (Lena et al., 2022). Previous studies, such as Leininger et al. (2019) recommended a 50 kPa as irrigation threshold for very fine sandy loam textured soil in Mississippi and Irmak et al. (2016) recommended a range from 20 kPa to 25 kPa for fine sandy textured soils and a range from 90 kPa to 110 kPa for silt loam and silty clay loam textured soils in Nebraska. However, these studies used SHPs based on either laboratory determinations (raw) using undisturbed soil cores (Lena et al., 2022) or estimated based on soil texture and pedotransfer functions using particle size fractions (Iqbal et al., 2020). These SHPs were not optimized for observed h or θ in actual field conditions to determine irrigation thresholds. Also, these studies did not consider significance of HCC, which is responsible to determine dynamic behavior of soil water movement. Irrigation scheduling using non-optimized SHPs does not account for the discrepancies between field observations of h or θ ; such discrepancies can also result in the determination of irrigation thresholds and amounts (Evelt et al., 2019).

Previous studies (Irmak et al., 2016; Leininger et al., 2019; Lena et al., 2022) did not consider soil water dynamics in determining the irrigation thresholds. The SHPs obtained based on laboratory experiment (Lena et al., 2022) or soil texture (Iqbal et al., 2020) can show spatial variability in irrigation thresholds. These are "raw SHPs" and these raw SHPs do not capture soil water dynamics in field conditions accurately. The impact of evaluating optimized SHPs considering soil water dynamics on determining irrigation thresholds has never been studied, which can address the spatial variability in soil water dynamics in field conditions and can result in effective irrigation thresholds. In this study, we evaluated the impact of raw and optimized SHPs on determining irrigation thresholds in a delineated field and presented a novel approach of optimizing the zone-specific irrigation thresholds. This study investigated the difference between uniform irrigation and zone-specific irrigation that accounts for zone-specific SHPs (both SWRC and HCC) in a farmer's field. We hypothesized that zone-specific irrigation thresholds and amounts greatly vary with optimized SHPs in each zone. Since field measurements are practically onerous, time-consuming, and include uncertainty in the measurements, it makes sense to establish a modeling setup to optimize effective SHPs for zone-specific characterization for more accurate irrigation and agricultural water management. Therefore, for this investigation, we selected two different management zones of an irrigated cornfield. For each management zone, we optimized the SHPs using measured h with inverse HYDRUS-1D modeling coupled with the Shuffled Complex Evolution (SCE-UA) algorithm. Based on the optimized model, we compared the currently applied irrigation (uniform application) with many different scenarios to schedule irrigation.

2. Material and methods

2.1. Study area description

This research was carried out in a cornfield belonging to a farmer located (34° 43' 06.46" N and 87° 23' 13.53" W) in the Town Creek Watershed in the Tennessee Valley Region of North Alabama, USA. The study region is classified as subtropical humid (Cfa, Köppen climate classification) (Kottek et al., 2006) with average annual precipitation of about 140 cm. The growing season usually occurs from April to October (Kumar et al., 2021b) and distribution of total monthly precipitation over the 2018 and 2019 growing season is shown in Fig. 1. The annual average precipitation (1951–2017) over the growing period of April to August was 57 cm, and the total rainfall received during this period of

the 2018 and 2019 growing seasons was 61 cm and 60 cm, respectively. Also, a recent study in the Southeast USA showed a 14 % increment in the annual average of precipitation by 2059, which would alter the time, erosivity characteristics, and distribution of the precipitation events (Takhellambam et al., 2021, 2022). The distribution of precipitation is very important over the growing season and has an impact on irrigation frequency and consequently on crop water demand. Therefore, if the precipitation is received during a high water demand period, irrigation amount and frequency will be decreased, and if the precipitation is received at the beginning or near the end of the growing season, it does not have much impact on irrigation. It can be observed that some of the months during the growing seasons have received minimum precipitation in 2018 and 2019. For example, July in 2018 and May-June in 2019 received the least precipitation over the growing season. Also, during the high water demand period from mid-May to mid-June 2019, the total precipitation received was only 3 cm. Although crops might receive enough precipitation at the end of the growing season, the time of precipitation is critical to reducing the crop water stress. Therefore, it is important to plan irrigation according to the crop water demand. The study field of 120 ha was equipped with a remotely controlled 625 m long Reinke center pivot irrigation system (Reinke Manufacturing Co., Inc., Deshler, NE, USA) with a mid-elevation spray application (MESA) sprinklers fitted with pressure regulators. The uniformity testing of the center pivot irrigation system was performed before the growing season with the catch can test (Dukes and Perry, 2006; Irmak et al., 2011; Ortiz et al., 2021). Readers can follow Ortiz et al. (2021) for detailed and stepwise procedure. Irrigation was triggered when there was a deficit of about 2.54 cm (as the upper limit) of irrigation depth in the root zone

compared with field capacity. The reason to use 2.54 cm as the upper limit was based on the maximum capacity of the center pivot irrigation system for applying irrigation in the field (Kumar, 2022a; Kumar et al., 2022b). The total irrigation amount applied by farmers in the field was 6.6 cm and 8.76 cm in 2018 and 2019 growing seasons, respectively. Corn (*Zea mays* L.), cotton (*Gossypium hirsutum* L.), soybean (*Glycine max* (L.) Merr.), and winter wheat (*Triticum aestivum* L.) are the major crops in the region (Kumar et al., 2021b).

The field was delineated in three irrigation management zones (zone 1, zone 2, and zone 3; however, we used zone 1 and zone 2 in this study due to contrasting properties in these zones) using the Management Zone Analyst (MZA, USDA-ARS, Columbia, MO, USA) software (Fridgen et al., 2004) based on terrain elevation data collected using a StarFire™ 6000 real-time kinematic GPS receiver (Deere and Company, Moline, IL, USA), historical yield data collected using yield monitors mounted on a harvester, and soil texture. A similar approach was adopted to delineate the management zones in cornfields of North Alabama (Kumar et al., 2022b), Central Alabama (Filho et al., 2020), and South Alabama (Jiménez et al., 2020). For soil texture, a disturbed soil sampling campaign was conducted from several locations for soil characterization in the study field at three depths 0–15 cm, 15–30 cm, and 30–60 cm using the hydrometer method (Bouyoucos, 1962) (Table 1). The slope was generated using terrain elevation data and it varied from 0 % to 10 %. A topographic wetness index (TWI) was also generated (Beven and Kirkby, 1979), and TWI varied from 5.11 to 18. Two contrasting zones (namely, zone 1 and zone 2) differing in slope, soil, TWI, and crop yield were selected for this study (Table 1). The zone 1 had lower slope, greater TWI, and greater crop yield as compared to zone 2, which had

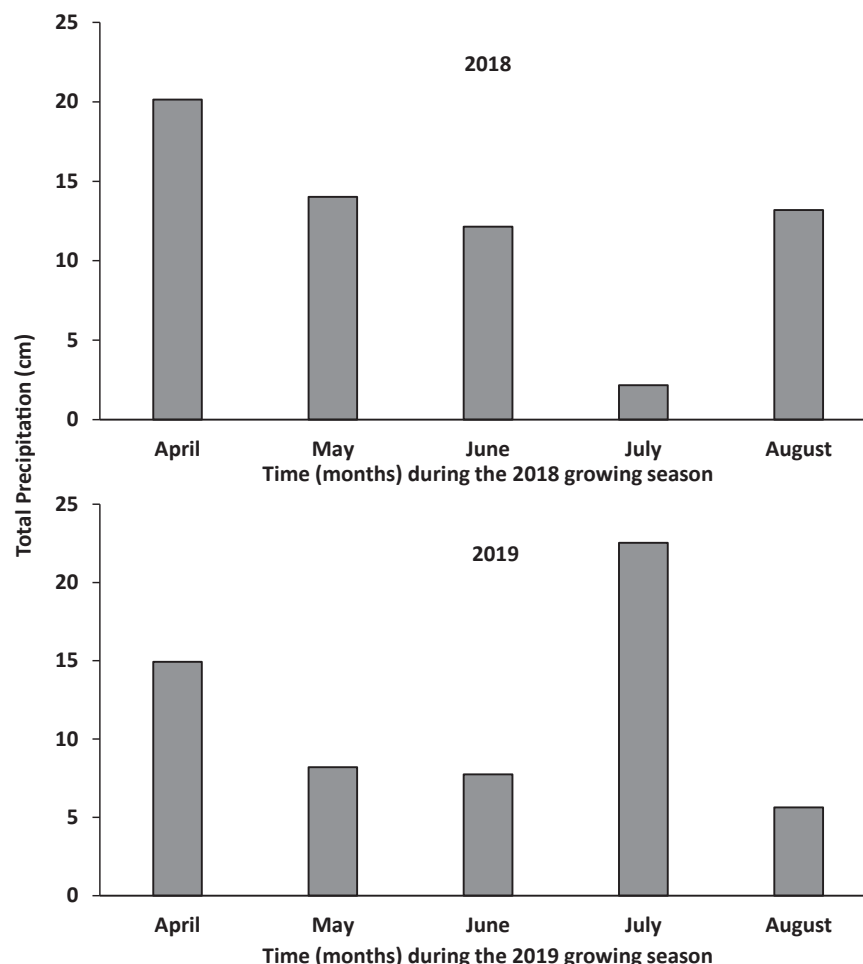


Fig. 1. Distribution of precipitation over the growing seasons.

Table 1

Zone-specific soil properties in the crop field. (* indicates data retrieved from SSURGO 2022).

Zones ^a	Depth (cm)	Sand (%)	Silt (%)	Clay (%)	Texture	Soil Series*	Soil Family*
Zone 1	0–15	17.5	42.3	40.2	Silty Clay	Abernathy-Emory	Fine-silty, siliceous, active, thermic Oxyaquic Hapludults
	15–30	17.6	50.1	32.3	Silty Clay Loam		
	30–60	12.9	51.4	35.7	Silty Clay Loam		
Zone 2	0–15	31.9	32.8	35.3	Clay Loam	Decatur	Fine, kaolinitic, thermic Rhodic Paleudults
	15–30	30.1	29.6	40.3	Clay		
	30–60	29	24.4	46.8	Clay		

^a The zone 1 and zone 2 are delineated irrigation management zones within the field.

greater slope, lower TWI, and lower crop yield within the field. Zone 3 had properties between zone 1 and zone 2, and, therefore, it was not used in the study. Since about 95 % of corn roots are distributed between 0 and 60 cm soil layer (Zhou and Zhao, 2019), an array of three Watermark sensors (Irrometer Co., Riverside, CA, USA) at multiple locations were installed in each zone to monitor zone-specific h at 0–15 cm, 15–30 cm, and 30–60 cm soil depths during the growing season. Watermark sensors are electrical resistance devices to monitor h in different soil types in crop fields. Sensors were installed on 01 May 2019 and removed on 11 August 2019. The corn was planted on 28 March 2019 and harvested on 30 August 2019. During the 2018 growing season, corn was planted on 10 April 2018 and harvested on 03 September 2018. An automatic weather station (Vantage Pro2 Plus, Davis Instruments, Hayward, CA, USA) was installed near field to record meteorological parameters (Kumar et al., 2021b). The groundwater depth below the surface is variable in the region, however, the annual average groundwater depth is 9 m from the nearest well in the region (USGS well: 343940087290601, https://nwis.waterdata.usgs.gov/usa/nwis/gwlevels/?site_no=343940087290601).

2.2. HYDRUS-1D numerical approximate model description

A numerically approximate one-dimensional model HYDRUS-1D was selected to simulate soil water dynamics in both zones during the growing seasons. HYDRUS-1D has been used as a reliable tool to model water dynamics of variably saturated soils in various field and laboratory conditions (Mokari et al., 2019; Šimůnek et al., 2016; Wyatt et al., 2021). HYDRUS-1D has been used in different scenarios such as rainfed irrigation (Iqbal et al., 2020), furrow irrigation (Tafteh and Sepaskhah, 2012), uniform irrigation (Fontanet et al., 2020), and root water uptake studies (Šimůnek and Hopmans, 2009; Zheng et al., 2017; Wang et al., 2021).

The model solves the governing Richard's equation by applying a finite element code. The one-dimensional equation for water flow in an unsaturated flow medium is described by Eq. (1).

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z} \left[K \left(\frac{\partial h}{\partial z} + 1 \right) \right] - S \quad (1)$$

where θ is soil water content [$L^3 L^{-3}$]; h is soil water pressure head [L]; t is time [T]; z is the vertical spatial coordinate [L]; K is unsaturated hydraulic conductivity [$L T^{-1}$]; and S is sink term in the water flow equation accounting for water uptake by roots [$L^3 L^{-3} T^{-1}$].

The zone-specific SHPs were modeled using van Genuchten (1980):

$$\theta(h) = \begin{cases} \theta_r + \frac{\theta_s - \theta_r}{[1 + (\alpha|h|)^n]^m}; & h < 0 \\ \theta_s; & h \geq 0 \end{cases} \quad (2)$$

$$K(h) = \begin{cases} K_s S_e^l \left[1 - \left(1 - S_e^{\frac{1}{m}} \right)^m \right]^2; & h < 0 \\ K_s; & h \geq 0 \end{cases} \quad (3)$$

$$S_e = \frac{\theta - \theta_r}{\theta_s - \theta_r} \quad (4)$$

θ_r is residual soil water content [$L^3 L^{-3}$]; θ_s is saturated soil water content [$L^3 L^{-3}$]; α [L^{-1}], m [-], and n [-] are shape parameters, where $m = 1 - 1/n$; l [-] is shape parameter in hydraulic conductivity function; K_s saturated hydraulic conductivity [$L T^{-1}$]; S_e is relative saturation or effective water content [-].

2.3. HYDRUS-1D model input parameters

2.3.1. Determination of zone-specific SHPs

For each zone, undisturbed soil core was collected using a sampler ring of stainless steel (5 cm in height and 8 cm in diameter). The sampler rings were carefully removed by digging out surrounding soils and stored in a container for transportation. The sample cores were saturated for 24 h before the soil water retention measurements using the evaporation method with the HYPROP measurement system (Peters et al., 2015; Peters and Durner, 2008). HYPROP measurements are limited by the cavitation point of the two tensiometers, and for this reason, water retention curves can be estimated from saturation until around pF 3. The pF is a logarithmic function of absolute h in cm. To determine the dry range of SWRCs, we used a dew-point hygrometer (WP4C Potentiometer, Meter Group, Pullman, WA, USA). For detailed information about developing SHPs, readers are suggested to follow Lena et al., (2021, 2022).

$$pF = \log_{10}(|h|) \quad (5)$$

2.3.2. Dynamic field measurements

To input meteorological data, the surface water flux (precipitation and irrigation), minimum and maximum temperatures, humidity, solar radiations, and wind speed were collected from the installed weather station. Irrigations applied by farmer during the 2018 and 2019 growing seasons were recorded in addition to precipitation for water input data in the model. The growth stages were recorded for corn to determine the associated crop coefficient during the growing season (Irmak, 2017). The plant heights and leaf area index (LAI) were also recorded in both zones during the field visits. The LAI was measured using an LAI-2200 Plant Canopy Analyzer (LI-COR Environmental, Lincoln, NE, USA) on the 57th, 72th, 92th and 128th day after planting during the growing season. The LAI was measured from four different locations within each zone with five replications for each location (Morata, 2020).

2.3.3. Initial and boundary conditions

The initial conditions were defined in terms of h in the model. Since this study is a field study, and the surface layer was subjected to the atmosphere, we defined the upper boundary condition (BC) as atmospheric BC with surface runoff in the HYDRUS-1D model. The lower BC was set to free drainage boundary at 100 cm soil depth, indicating no interaction with the groundwater. A spatial discretization of 0.5 cm, leading to 201 nodes across the entire soil profile, and no hysteresis in soil water retention was considered. The observed nodes (top, center, and bottom) were defined at the same depths of 15 cm, 30 cm, and 60 cm, where Watermark sensors were installed in both zones.

2.3.4. Root water uptake

HYDRUS-1D simulates root water uptake (RWU) of a crop during the growing season. The RWU, i.e., the sink term, S , in Richard's equation, was computed using an equation given by Feddes et al. (1978).

$$S = \alpha(h)\beta(z)T_p \quad (6)$$

where $\alpha(h)$ [-] is a dimensionless response function of h ($0 < \alpha(h) < 1$) which accounts for RWU stress; $\beta(z)$ is normalized water uptake distribution [L^{-1}]; T_p is a potential transpiration rate [LT^{-1}].

Water stress causes RWU reduction, $\alpha(h)$, can be explained as

$$\alpha(h) = \begin{cases} 0; h > h_1 \text{ or } h < h_4 \\ \frac{h - h_1}{h_2 - h_1}; h_1 > h > h_2 \\ 1; h_2 \leq h \leq h_3 \\ \frac{h - h_4}{h_3 - h_4}; h_3 > h > h_4 \end{cases} \quad (7)$$

where h_1 is called anaerobiosis point, i.e., water uptake considered to be zero close to the saturation ($h > h_1$); h_4 is pressure head at the wilting point and water uptake is also considered to be zero when $h < h_4$. Maximum RWU happens between h_2 and h_3 , and it decreases or increases linearly between the pressure heads h_3 and h_4 or h_1 and h_2 , respectively (Fig. 2). Since 95 % of the corn roots are linearly distributed within the 0–60 cm soil profile (Zhou and Zhao, 2019), and for this reason, we assumed linear root distribution function to a depth of 60 cm. The parameters in Wesseling (1991) were selected for the root water uptake of corn in both zones.

2.4. Inverse modeling to optimize zone-specific SHPs- HYDRUS-1D a reference scenario

We used inverse modeling to optimize a single layer (0–100 cm) effective SHPs using h measured at three depths (0–15 cm, 15–30 cm, and 30–60 cm) in both zones. Due to fairly uniform soil layers and textural homogeneity within the soil depths in each zone, we used single-layer SHPs. We converted h into pF to perform inverse modeling in HYDRUS-1D. The HYDRUS-1D model was coupled with the SCE-UA developed by Duan et al. (1994) to perform optimization of SHPs (Diamantopoulos et al., 2012; Qu et al., 2014). An objective function (Obj) was defined for minimization in the SCE-UA algorithm.

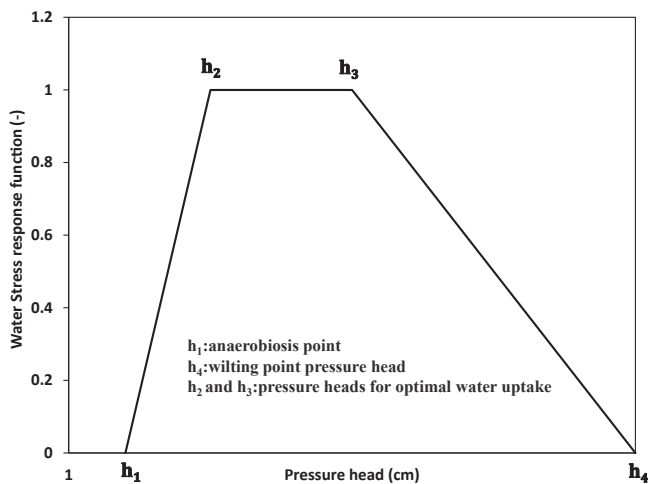


Fig. 2. Schematic diagram of root water uptake stress response function. h_1 is anaerobiosis point, h_4 is the wilting point pressure head, and h_2 and h_3 are pressure heads where water uptake was considered optimal.

$$Obj = \sum_{j=1}^{nn} [pF_j - \widehat{pF_j}(x)]^2 \quad (8)$$

where $pF_j = [pF_{15cm}, pF_{30cm}, pF_{60cm}]$ is a vector of daily observations of h as pF values; $\widehat{pF_j} = [\widehat{pF}_{15cm}, \widehat{pF}_{30cm}, \widehat{pF}_{60cm}]$ a vector that contains daily HYDRUS-1D predictions of h as pF values; $x = (\theta_r, \theta_s, \alpha, n, K_s, l)$ is a vector of soil hydraulic parameters used in van Genuchten model, j is the number of each observation point, and nn is the total number of measurements at each soil depth. The model was set up from 05 April 2018–30 August 2019 until harvesting the crop. From 05 April 2018 until 30 April 2019, the model was trained to set initial conditions to run the simulations for the optimization of SHPs.

2.5. Goodness of fit

Different indices of goodness of fit have been calculated: root mean square error (RMSE), index of agreement (IA), and relative root mean square error (RRMSE) were used in this study.

$$RMSE = \sqrt{\frac{1}{nn} \sum_{j=1}^{nn} (\widehat{pF_j} - pF_j)^2} \quad (9)$$

$$IA = 1 - \frac{\sum_{j=1}^{nn} (\widehat{pF_j} - pF_j)^2}{\sum_{j=1}^{nn} (|\widehat{pF_j} - pF_{avg}| + |pF_j - pF_{avg}|)^2} \quad (10)$$

$$RRMSE = \frac{\sqrt{\frac{1}{nn} \sum_{j=1}^{nn} (\widehat{pF_j} - pF_j)^2}}{\sum_{j=1}^{nn} (pF_j)} \quad (11)$$

where nn is the total number of observations; $\widehat{pF_j}$ and pF_j are simulated and observed h as pF values, respectively, and pF_{avg} is average h as pF values. The lower the RMSE value the better is the model performance in predicting the data in terms of its absolute error. IA value ranges from 0 to 1, and close to 1 indicates a better performance of the model. RRMSE indicates the performance of the model excellent when $RRMSE < 0.1$, good if $0.1 \leq RRMSE \leq 0.2$, fair if $0.2 < RRMSE \leq 0.3$, and poor if $RRMSE > 0.3$ (Wang et al., 2018).

2.6. Irrigation thresholds and scheduling

We adopted a 35 % SWD often recommended for most agronomic crops, especially corn (Girona et al., 2002; Irmak et al., 2016; Liang et al., 2016) to compute irrigation thresholds (Allen et al., 1998; Leininger et al., 2019). The thresholds at 35 % SWD can be determined using the following equations.

$$|h|_{SWD} = \frac{\left\{ \left[\frac{\theta_s - \theta_r}{\theta_{SWD} - \theta_r} \right]^{1/m} - 1 \right\}^{1/n}}{\alpha} \quad (12)$$

$$\theta_{SWD} = \theta_{FC} - [SDW(\theta_{FC} - \theta_{PWP})] \quad (13)$$

$$I_{SWD} = \sum (\theta_{FC} - \theta_{SWD})Z \quad (14)$$

where $|h|_{SWD}$ is irrigation threshold value as h at 35 % SWD [L]; θ_{SWD} is soil water content at 35 % SWD [$L^3 L^{-3}$]; θ_{FC} is soil water content at field capacity [$L^3 L^{-3}$]; θ_{PWP} is soil water content at permanent wilting point [$L^3 L^{-3}$]; I_{SWD} irrigation amount or depth [L]; Z is thickness of soil layer [L]. Field capacity at 33 kPa and permanent wilting point at 1500 kPa were used to determine irrigation thresholds and schedule irrigation in this study.

The precipitation was the only water input variable that was kept in

the model. We defined these models as No Irrigation models (NI). The trigger irrigation module within the HYDRUS-1D model was selected to schedule the irrigations using the determined irrigation thresholds and amounts at 35 % SWD with zone-specific optimized SHPs. The irrigation scheduling was performed using the observation nodes at 15 cm (node 1), 30 cm (node 2), and 60 cm (node 3), as the hypothetical depth of an installed sensor. At each observation node, multiple irrigation scenarios were created in each zone as summarized in Table 2. We triggered the irrigations on the same day without delay (0Lag), after one day (1Lag), after two days (2Lag), after three days (3Lag), after four days (4Lag), and after five days (5Lag) when irrigation threshold reached (Table 2). The idea of different irrigation scheduling was to bring the models of irrigation scenarios to the reference model status and reduce water stress or meet potential water requirements (Ramos et al., 2009).

2.7. Actual root water uptake and irrigation water for various irrigation scenarios

We examined actual root water uptake (ARWU) and irrigation applied by different scenarios (Zheng et al., 2017; Zhou and Zhao, 2019). The HYDRUS-1D model tries to meet every day potential root water uptake (PRWU), and if the water is not available, stress function reduces it to ARWU. The ratios of ARWU and PRWU for various scenarios were calculated to estimate the water stress in both zones. A value close to one shows no water stress in the crop, and a value close to zero indicates severe water stress in the crops (Ramos et al., 2009) and inefficient irrigation. Increasing ARWU helps increase crop yield and loss in ARWU results in yield reduction (Ramos et al., 2011).

3. Results and discussion

3.1. Inverse HYDRUS-1D model performance for optimizing SHPs

A single porosity model characterized the SHPs to satisfy the relation for both zones using HYPROP-FIT software (Mualem, 1976; van Genuchten, 1980). The SHPs in zone 1 and zone 2 obtained from HYPROP-FIT software using the data collected from HYPROP measurement system are shown in Fig. 3 and these SHPs were “raw SHPs”. These raw SHPs were used as input parameters in the inverse HYDRUS-1D model to start the simulations for optimizing zone-specific SHPs. The raw SHPs showed the spatial variability between zone 1 and

zone 2 (Fig. 3). Both zones had distinct patterns of SHPs, which means invariable SHPs cannot be assumed throughout the field for irrigation management. It was inferred from the raw SHPs that considering uniform SHPs within the field can be a flawed irrigation strategy due to zone-specific variability. Assuming uniform SHPs will trigger irrigation at one threshold and one rate in the entire field, which can result in over-irrigation in some parts and under-irrigation in others. It was also implied from these raw SHPs that we need to consider spatial variability between the zones for better irrigation management and adopt zone-specific SHPs. The raw SHPs did not consider the soil water dynamics within the soil profiles to address the significance of the observed h in each zone, and ignored the soil water movement for determining the effective irrigation thresholds. Although raw SHPs showed the importance of zone-specific SHPs between the zones, the raw SHPs were not able to simulate field observations of h measured during the growing season. This reason signified the need for optimizing zone-specific SHPs to simulate the soil water dynamics in each zone using field observations.

Inverse solution using measured h indicated the importance and need for optimizing the zone-specific SHPs (Fig. 3) in precision agriculture, for zone-specific irrigation improvements across the field to conserve agricultural water, and for the adoption of VRI technology. The optimized SHPs obtained using inverse HYDRUS-1D modeling were different from raw SHPs obtained from HYPROP-FIT software in the lab. Using the inverse solution in each zone, we found more effective zone-specific SHPs (Fig. 3), which simulated soil water dynamics with good agreement in both zones (Table 3). The simulated pF values fitted well with measured pF values for all the soil depths using optimized SHPs in both zones, as shown in Figs. 4 and 5.

The HYDRUS-1D model described relatively well the observed data in all three soil layers in both zones using a single layer zone-specific optimized SHPs. The variability in soil water dynamics were due to the attributes considered in delineating management zones, for example, topographic attributes and soil texture. At the beginning of simulations, the gap between measured and simulated data can be due to the stabilization issues associated with the Watermark sensors in the crop fields (Irmak et al., 2016).

According to model testing at multiple locations in zone 1, the model performed best at a 15–30 cm soil depth with the best agreement and minimum error between observed and simulated data, followed by 0–15 cm and 30–60 cm (Table 3). However, in zone 2, the observed and simulated data had the best agreement at 0–15 cm with the lowest error among all soil depths. In comparison between the two zones, zone 1 performed better than zone 2 since the statistical indices were better in zone 1. The variation in the model performances in both zones was due to differences in the SHPs (Fig. 3), differences in the topographic attributes (low slope and high TWI in zone 1 and high slope and low TWI in zone 2), and differences in root water uptake (Fig. 6), which affected the soil water dynamics in both zones (Figs. 4 and 5). It can also be noticed from the recorded plant height and LAI values, which were greater in zone 1 and lower in zone 2.

The one-layer optimized HYDRUS-1D model is a simplified version using single layer SHPs of the complex system, which can be used to simulate soil water dynamics at all three different layers (Figs. 4 and 5). However, comparing previous studies for HYDRUS-1D model performance using multilayer SHPs, Li et al. (2014) conducted a study in an irrigated rice field in China and the minimum average RMSE achieved was 2.61 cm using the individual retention curves for each layer from 0 to 100 cm at 20 cm increment depth obtained from retention curves (RETC) computer program. Wyatt et al. (2017) reported an RMSE ranged from 2.95 to 3.53 as pF higher in deep soil layer in a study to estimate groundwater recharge at four different sites in Oklahoma using the HYDRUS-1D model. However, these studies did not take all parameters of SWRC and HCC ($\theta_r, \theta_s, \alpha, n, K_s, I$) into account to find optimized SHPs. Some underpredictions and over predictions of simulated data compared with observed data during the simulation period can be

Table 2

Descriptions of HYDRUS-1D models created for irrigation scheduling in both zones.

Depth	Irrigation trigger description ^a	Notations for Irrigation Scheduling Models
15 cm	0Lag	Top Node 0Lag
	1Lag	Top Node 1Lag
	2Lag	Top Node 2Lag
	3Lag	Top Node 3Lag
	4Lag	Top Node 4Lag
30 cm	5Lag	Top Node 5Lag
	0Lag	Center Node 0Lag
	1Lag	Center Node 1Lag
	2Lag	Center Node 2Lag
	3Lag	Center Node 3Lag
60 cm	4Lag	Center Node 4Lag
	5Lag	Center Node 5Lag
	0Lag	Bottom Node 0Lag
	1Lag	Bottom Node 1Lag
	2Lag	Bottom Node 2Lag
	3Lag	Bottom Node 3Lag
	4Lag	Bottom Node 4Lag
	5Lag	Bottom Node 5Lag

^a 0Lag: Same day of threshold reached; 1Lag: After 1 day of threshold reached; 2Lag: After 2 days of threshold reached; 3Lag: After 3 days of threshold reached; 4Lag: After 4 days of threshold reached; 5Lag: After 5 days of threshold reached.

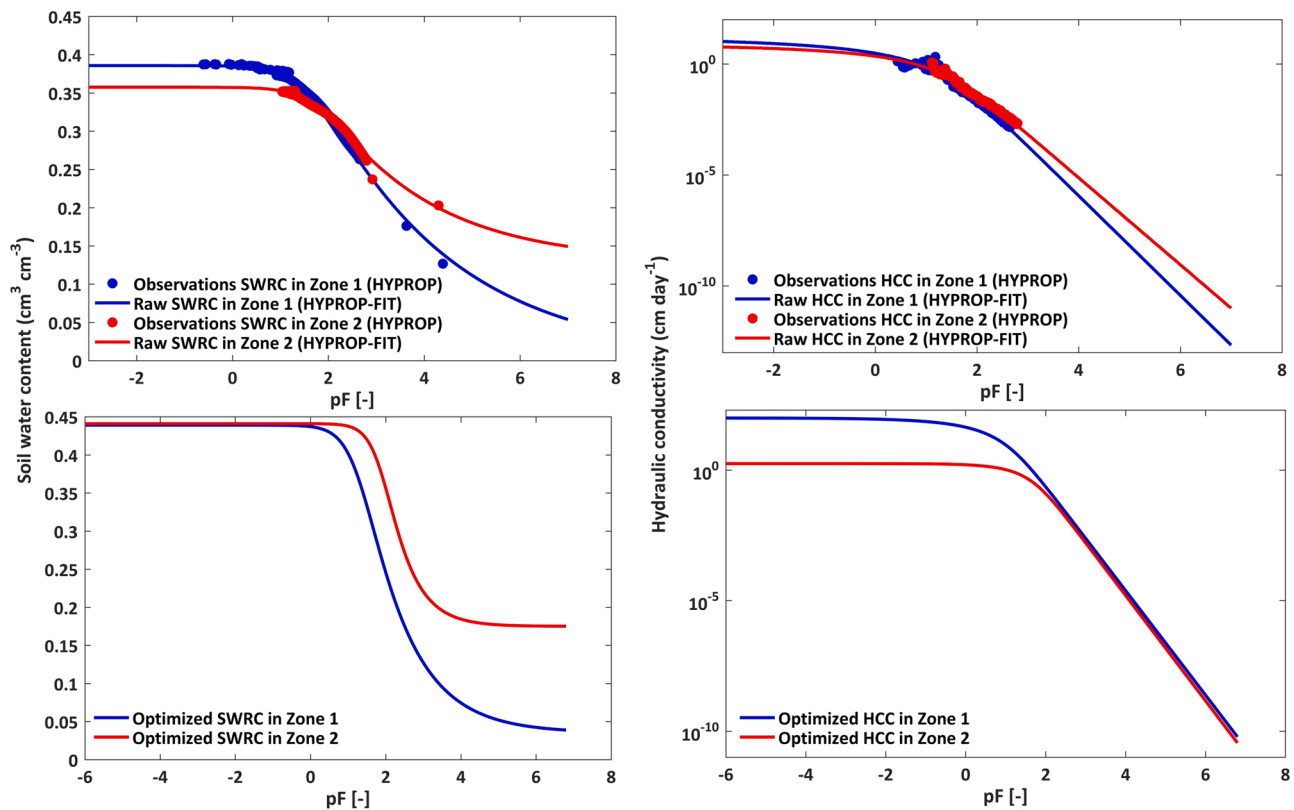


Fig. 3. Raw SHPs; SWRC (top left) and HCC (top right). The observations (dots) were obtained from HYPROP measurement system and fitted with HYPROP-FIT software. Zone-specific optimized SHPs; SWRC (bottom left) and HCC (bottom right) obtained from inverse modeling. The pF is a logarithmic function of absolute pressure head (h). The zone 1 and zone 2 are the delineated irrigation management zones within the field. SHPs: soil hydraulic properties, SWRC: Soil water retention curves, HCC: hydraulic conductivity curve.

Table 3

Average performance of inverse HYDRUS-1D model at each soil depth in both zones.

Zones ^a	Statistics ^b	0–15 cm	15–30 cm	30–60 cm
Zone 1	RMSE	0.36	0.29	0.38
	IA	0.82	0.84	0.78
	RRMSE	0.001	0.001	0.001
Zone 2	RMSE	0.33	0.38	0.40
	IA	0.83	0.79	0.71
	RRMSE	0.001	0.002	0.001

^a The zone 1 and zone 2 are delineated irrigation management zones within the field.

^b RMSE is root mean square error, IA is index of agreement, and RRMSE is relative root mean square error.

associated with soil water retention behavior of the fine-textured soil (Iqbal et al., 2020), hysteresis effect (Zhou and Zhao, 2019), and measurements errors associated with the sensors (Li et al., 2014; Mokari et al., 2019). Mokari et al. (2019) reported an IA that varied from 0.44 to 0.73. Discrepancies between observed and simulated data in the zones due to lateral leakage and macro-pores have also been reported in previous studies (Akay et al., 2008; Wang et al., 2011; Zhang et al., 2018). Based on the statistics to evaluate the goodness of fit, the model performance was evaluated as excellent to optimize zone-specific SHPs since our results were quite consistent and better than previous studies (Li et al., 2014; Mokari et al., 2019; Wyatt et al., 2017) using a single layer SHPs.

Fig. 6 shows the actual and potential RWU in both zones simulated using the optimized HYDRUS-1D model during the growing seasons of corn. During the seed germination stage, RWU was almost zero and increased with growing stages. The different patterns of RWU can be

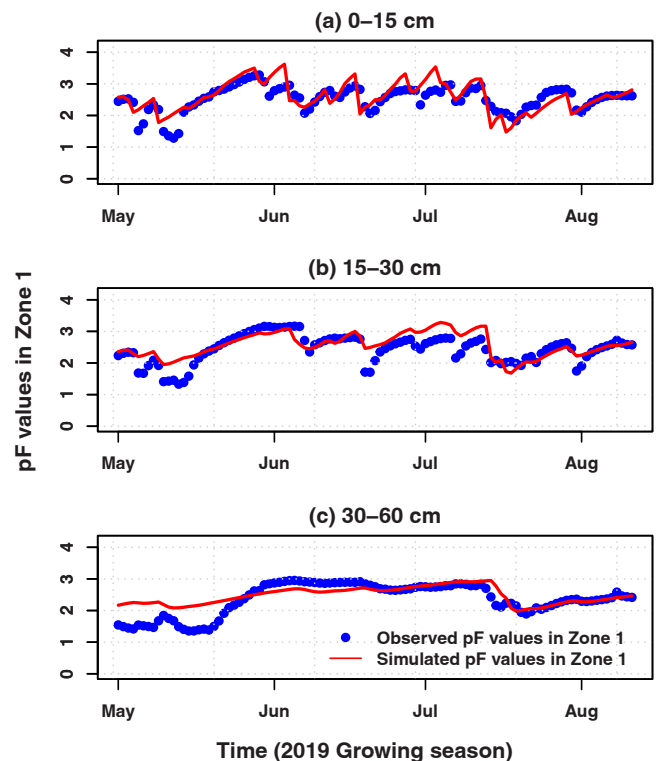


Fig. 4. Observed and simulated pF values in zone 1 at (a) 0–15 cm, (b) 15–30 cm, and (c) 30–60 cm soil depths during the 2019 growing season. The pF is a logarithmic function of absolute pressure head (h).

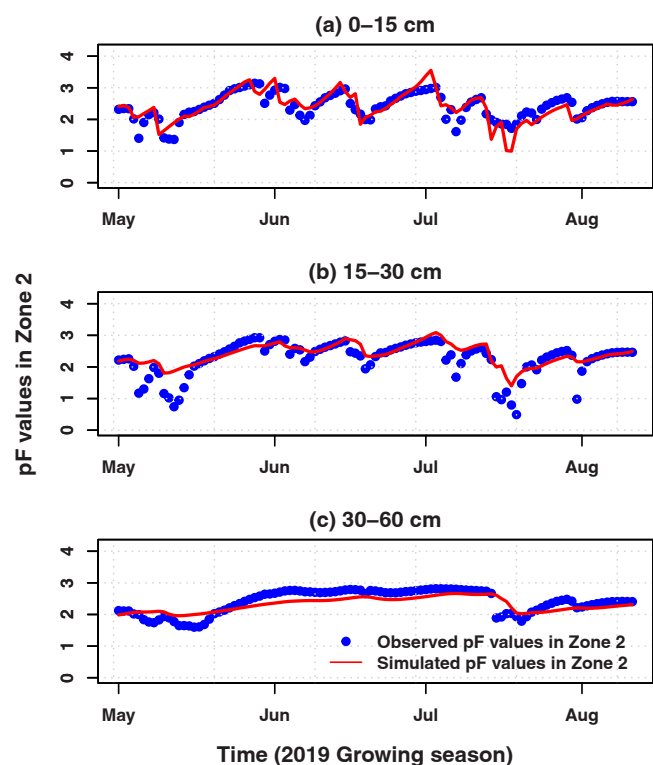


Fig. 5. Observed and simulated pF values in zone 2 at (a) 0–15 cm, (b) 15–30 cm, and (c) 30–60 cm soil depths during the 2019 growing season. The pF is a logarithmic function of absolute pressure head (h).

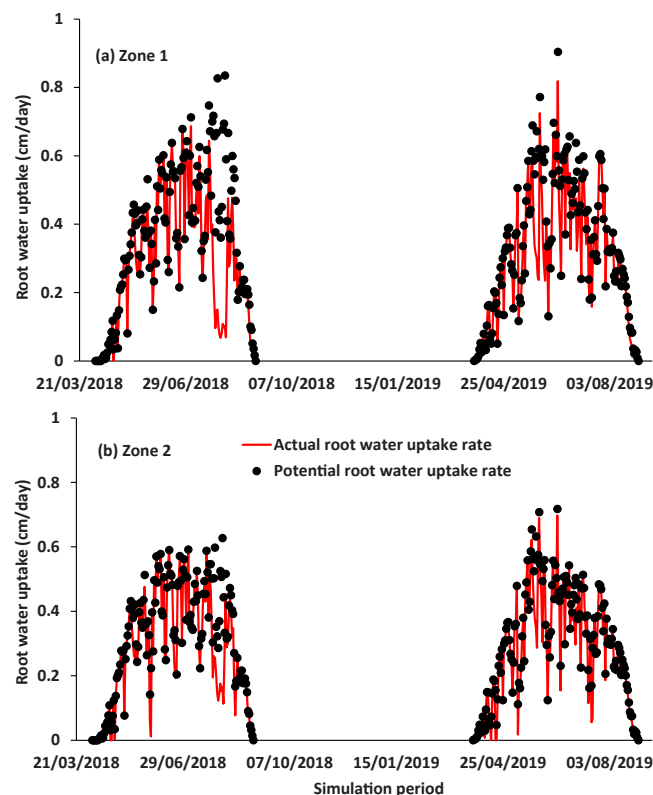


Fig. 6. Root water uptake in (a) zone 1 and (b) zone 2 during 2018 and 2019 growing seasons. The zone 1 and zone 2 are the delineated irrigation management zones within the field.

noticed in the 2018 and 2019 growing seasons in both zones. The actual RWU gradually increased and reached a maximum of 0.69 cm/day in zone 1 and 0.59 cm/day in zone 2 around the 86th day after planting in 2018, and 0.82 cm/day in zone 1 and 0.70 cm/day in zone 2 around the 80th day after planting in the 2019 growing season. The variability in RWU in zone 1 and zone 2 was due to differences in SHPs obtained after optimization, for example, the slow movement of water in zone 2 compared with zone 1. Additionally, the terrain properties of both zones indicated the low slope and high TWI in zone 1 and high slope and low TWI in zone 2. A higher TWI created the water accumulation conditions in zone 1, and a higher slope led to sub-surface lateral flow in zone 2, which affected the RWU variability in both zones along with the differences in SHPs (Kumar, 2022a). The variability in RWU between two zones affected the crop characteristics as a greater plant height and LAI in zone 1 than in zone 2. The RWU results are consistent with previous studies (Iqbal et al., 2020; Zhou and Zhao, 2019). The maximum RWU increased in both zones from 2018 to 2019 growing seasons. The RWU increased from 2018 to 2019 due to increased precipitation and increased irrigation in the field during the growing seasons. The zone 1 had silty clay loam texture in the soil profile; however, zone 2 had clay texture. The clay texture has maximum water holding capacity and minimum water releasing property (Çakir and Cangir, 2021). On the other hand, silt type soils have slightly less water holding capacity than clay, but have greater water releasing property than clay soils (Çakir and Cangir, 2019, 2021). Çakir and Cangir (2021) found the similar results with variable clay contents in the soils in Turkey. Also, plant available water (field capacity-permanent wilting point) in the soil profile of zone 1 was 1.6 times greater than zone 2. Therefore, the ARWU was greater in zone 1 than zone 2 in both seasons. This variation of RWU in both zones and the differences in zone-specific SHPs had impact on plant characteristics. For example, the maximum measured average LAI was $4.13 \text{ cm}^2 \text{ cm}^{-2}$ and $3.34 \text{ cm}^2 \text{ cm}^{-2}$ in zone 1 and zone 2, respectively on the 92nd day after planting during the 2019 growing season. On the same day of LAI measurements, the plant height was 225 cm and 195 cm in zone 1 and zone 2, respectively. These field measurements for plant characteristics explain the zone-specific differences in root water uptake and root water uptake is controlled by zone-specific SHPs. Similar observations have been found in previous studies (Aggarwal et al., 2017; Wang et al., 2016).

3.2. Determination of irrigation thresholds and amounts

The accurate determination of an irrigation threshold and amount is critical for triggering zone-specific irrigation in crop fields. Table 4 summarized the irrigation thresholds and amounts determined using zone-specific optimized SHPs. The irrigation thresholds were optimized to 76 kPa from a raw value of 102 kPa in zone 1 and 59 kPa from 69 kPa in zone 2. The corresponding irrigation amounts were optimized to 2.18 cm in zone 1 and 1.48 cm in zone 2. Optimizing zone-specific SHPs optimized the thresholds to trigger irrigation and reduced the irrigation amount to 2.18 cm in zone 1 and 1.48 cm in zone 2 (Table 4). Among a very few studies that report irrigation thresholds, Irmak et al. (2016) suggested a range of irrigation thresholds from 90 kPa to 110 kPa for silt loam to silty clay loam soils in Nebraska. However, the range was not estimated using optimized SHPs in field conditions, and therefore, did

Table 4

Irrigation thresholds and amounts using zone-specific optimized and raw soil hydraulic properties at 35 % soil water depletion.

Zones ^a	Irrigation threshold		Irrigation amount	
	Optimized	Raw	Optimized	Raw
Zone 1	76 kPa	102 kPa	2.18 cm	2.6 cm
Zone 2	59 kPa	69 kPa	1.48 cm	1.9 cm

^a The zone 1 and zone 2 are the delineated irrigation management zones within the field.

not hold site-specific physical and hydraulic characteristics. Generally, farmers apply irrigation uniformly based on the feel of the soil or looking at the crop during the growing season (Kumar et al., 2021b). Delineating the irrigation management zones and adopting zone-specific irrigation strategies can help avoid water stress and water excess in the crop field. Irrigation threshold determined using raw SHPs can mislead the time and amount to trigger the irrigation since it did not simulate soil water dynamics within the zones. Optimizing the irrigation thresholds to trigger irrigation can help consider effective irrigation rate, time, and amount during the growing season.

3.3. Irrigation scheduling and actual root water uptake with different irrigation scenarios

The zone-specific determined thresholds and amounts using optimized SHPs were adopted in the HYDRUS-1D model to develop scenarios of irrigation scheduling. Among all different irrigation scenarios, scenarios with 0Lag, 1Lag, 2Lag, 3Lag, 4Lag, and 5Lag using the top node (15 cm) and center node (30 cm) in zone 1 reduced the water stress conditions and achieved an ARWU/PRWU ratio greater or equivalent to the reference scenario (Fig. 7) for both 2018 and 2019 growing seasons. The actual irrigations applied in the field were 6.6 cm in 2018 and 8.76 cm in 2019, which were input in the reference model. The reference model had the ARWU/PRWU ratios of 0.8 in 2018 and 0.9 in 2019 (Fig. 7). Among all the scenarios, the maximum irrigation amount applied was 21.8 cm in 2018 and 19.62 cm in 2019 with 0Lag and 1Lag using the top node, and achieved an ARWU/PRWU ratio greater than 0.97 in both seasons. However, the minimum irrigation applied was 10.9 cm in 2018 with 5Lag and 6.54 cm in 2019 with 0Lag, 1Lag, 2Lag, and 3Lag using the bottom node (supplementary Fig. S1), and achieved a minimum ARWU/PRWU ratio of 0.83 in both seasons. The distribution of precipitation over the growing seasons is the most critical component to decide the irrigation (Filho et al., 2020). Based on total precipitation distribution over the growing seasons, the maximum precipitation occurred in the beginning and near the end of growing seasons (Fig. 1). Therefore, looking at the actual irrigation data applied in the field, 71 % of actual irrigations were triggered during July 2018 and 100 % in May-June 2019. The reference model had ARWU/PRWU ratios of 0.7 in July 2018 and 0.8 in May-June 2019. Out of total irrigations applied by

the top and center nodes, more than 61 % was triggered in July 2018 and more than 75 % was triggered in May-June 2019. If no irrigation was applied during these periods, the crop was water-stressed with an ARWU/PRWU ratio of 0.44 in July 2018 and 0.68 in May-June 2019 due to lack of precipitation in this period (Fig. 1). As an ideal case to adopt an irrigation scenario, the top and center nodes with 5Lag improved the RWU in plants and performed better than the reference model in 2018 and equivalent to reference model in 2019 in zone 1. The impact of lack of precipitation over the growing seasons and the need for irrigation can be understood with an example of the top node with 5Lag irrigation scenario. The top node with 5Lag scenario applied 15.26 cm and 13.08 cm as total irrigation in 2018 and 2019 growing seasons, respectively (supplementary Fig. S1), which was 8.6 cm and 4.26 cm of excess irrigation in 2018 and 2019, respectively. Applying this excess irrigation water increased the RWU and improved ARWU/PRWU ratio. Out of 15.26 cm and 13.08 cm of irrigation water, 8.72 cm was applied in July 2018 and in May-June 2019. It was found that excess irrigation (8.6 cm in 2018 and 4.26 cm in 2019) contributed to the these periods of less precipitation in both growing seasons. Applying this excess irrigation water improved the ARWU/PRWU ratio in the water stress months (July 2018 and May-June 2019) and entire growing season (Fig. 7). The top and center nodes in zone 1 can perform well to monitor h and to trigger irrigation with any lag scenario using HYDRUS-1D. Irrigation scenarios using the bottom node, which was at the 60 cm in zone 1 increased the ARWU and improved ARWU/PRWU ratio (Fig. 7) for all irrigation scenarios compared with the reference model in 2018. However, the bottom node achieved a minimum ARWU/PRWU ratio of 0.83 in 2019 (Fig. 7). The bottom node triggered minimum irrigation in both growing seasons (supplementary Fig. S1) since soil water pressure did not reach irrigation thresholds frequently. Using the bottom node to trigger irrigation, greater than 60 % of the irrigation amount was applied in July, and more than 67 % was applied May-June of 2019.

In all the irrigation scenarios using top, center, and bottom nodes, RWU was mostly affected in the middle of growing seasons, where minimum precipitation occurred. This middle period of growing seasons was critical for the plants, which need maximum water uptake for plant growth and better yield production. A similar effect of precipitation distribution and the need for irrigation were found by Filho et al. (2020) in Alabama and Irmak et al. (2012) in Nebraska. Also, the excess

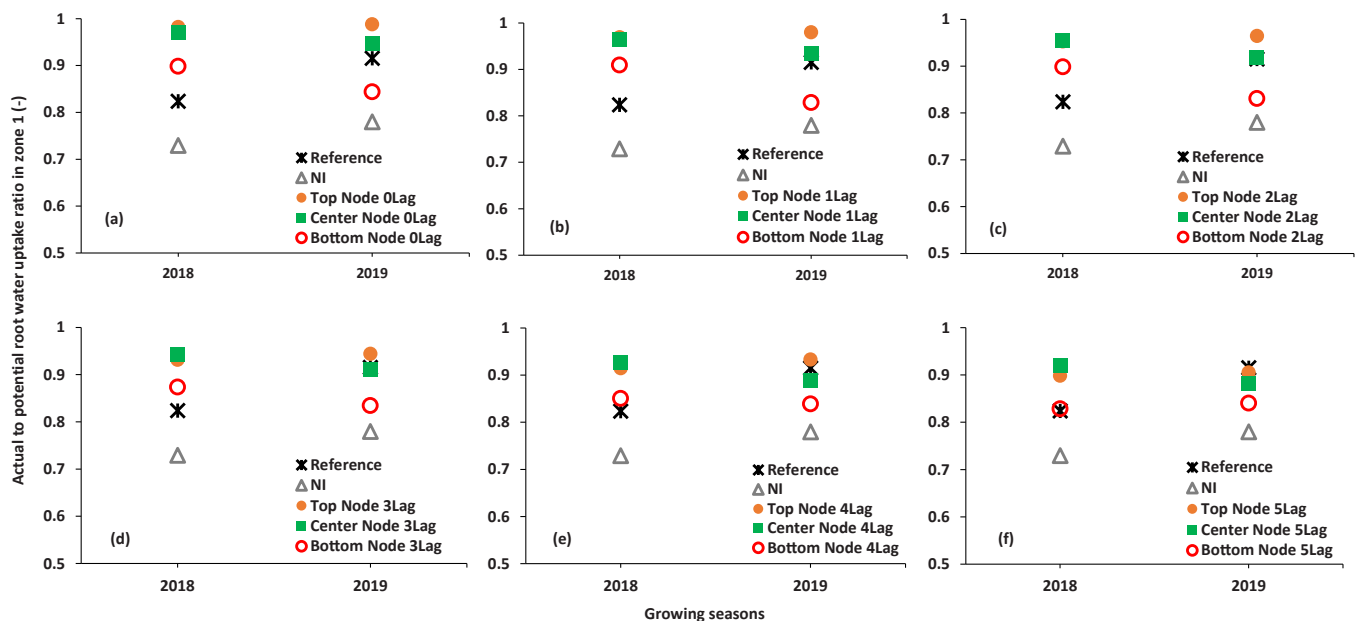


Fig. 7. Actual to potential root water uptake ratios in zone 1 for (a) 0Lag, (b) 1Lag, (c) 2Lag, (d) 3Lag, (e) 4Lag, and (f) 5Lag irrigation scenarios during 2018 and 2019 growing seasons. Reference scenario is the inverse model and NI is the model without irrigation data. Zone 1 is the delineated irrigation management zone within the field.

irrigation occurred during the water stress period of growing season (July 2018 and May–June 2019), which increased the ARWU and consequently ARWU/PRWU ratios. Although farmers trigger irrigation based on the feel of the soil and looking at the crop, they end up with over-irrigation in some areas and under-irrigation in others. It will be wise to adopt an irrigation strategy and install the sensors at 15 cm or 30 cm in zone 1 since these depths showed an increased ARWU/PRWU ratio than the reference scenario.

In zone 2, the top node in 2018 and 2019 (except 4Lag and 5Lag in 2019) and center node in 2018 increased the ARWU, reduced the water stress, and produced a greater ARWU/PRWU ratio (Fig. 8) as compared with the reference model for all irrigation scenarios in zone 2. The center node in zone 2 increased ARWU and achieved an ARWU/PRWU ratio greater or equivalent to the reference model (Fig. 8) for all scenarios of 2018, but not in 2019. These models showed improved RWU and reduced the water stress conditions in plants in 2018 growing season. The minimum value of ARWU/PRWU ratio using center node to trigger irrigation was 0.86 in 2019 for 3Lag, 4Lag, and 5Lag scenarios. However, the bottom node did not practice irrigation in 2019 and triggered less irrigation than the reference scenario in 2018 except 0Lag and 1Lag (supplementary Fig. S2). The bottom node did not improve the RWU and produced the ARWU/PRWU ratio equivalent to the NI model in both growing seasons (Fig. 8). It was inferred that the bottom node (60 cm) cannot be used to monitor h and cannot be used to trigger the irrigation in zone 2.

As an ideal case to adopt an irrigation scenario in zone 2, the top node with 1Lag and 2Lag improved the RWU in both seasons and performed better than the reference model. These scenarios improved the ARWU/PRWU ratio to 0.96 in 2018 and 0.94 in 2019 (supplementary Fig. S2) and reduced the water stress in the crop. Irrigation scenario using top node with 3Lag had better ARWU/PRWU ratio in 2018 and equivalent ratio in 2019 compared with the reference scenario. The maximum applied irrigation was 13.32 cm with the center node (supplementary Fig. S2) using the 0Lag and 1Lag scenario in 2018. However, minimum irrigation was 0 cm using the bottom node (supplementary Fig. S2). Similar to zone 1, most of the irrigations in zone 2 were triggered in July 2018 and in May–June 2019 due to the lack of precipitation over the growing seasons (Fig. 1). Irrigation scenarios using the top and

center nodes triggered greater than 73 % of the total irrigation in July 2018 and greater than 72 % in May–June 2019. Without irrigation, NI scenarios had the ARWU/PRWU ratio of 0.58 and 0.76 in July 2018 and in May–June 2019, respectively, similar to the bottom node scenarios, which showed the water stress and lack of water in zone 2. For example, the top node with the 4Lag scenario applied 11.84 cm as the total irrigation in both growing seasons (supplementary Fig. S2), which was 5.24 cm and 3.08 cm of excess irrigation in 2018 and 2019, respectively, as compared with reference scenario. The applied excess irrigation increased the RWU and improved the ARWU/PRWU ratio. Out of 11.84 cm of irrigation water, 7.4 cm was applied in July 2018 and 8.88 cm in May–June 2019. It was found that excess irrigation (5.24 cm in 2018 and 3.08 cm in 2019) contributed to these periods of less precipitation in both growing seasons. Irrigation scheduling using the top node performed better or equivalent to the reference model for various irrigation scenarios during the growing seasons in zone 2. Therefore, farmers can adopt an irrigation strategy in zone 2 by installing sensors at 15 cm soil depth to avoid water excess or severe water stress conditions during the growing season.

The model can achieve maximum RWU and reduce the water stress in both seasons. Also, assuming one irrigation threshold or using laboratory-drawn SHPs for the entire field can be a flawed irrigation management practice. Optimizing zone-specific SHPs can avoid discrepancies in field observation and result in better irrigation practice. Our results showed that each zone should be managed differently to improve water use and root water uptake. The results showed that optimizing irrigation thresholds can reduce the irrigation amount to be applied and simultaneously can increase the ARWU/PRWU ratio. However, if the calculated irrigation depth in any zones is greater than the irrigation system capacity or the application rate exceeds the infiltration rate, the irrigation depth needs to be reduced while the frequency to irrigate should increase. Reduction in irrigation depth and increased frequency can simultaneously be achieved by adopting a lower threshold of SWD. Reducing the irrigation depth will provide more freedom to the plant for water uptake and increase the ARWU/PRWU ratio during the growing season. The lack of irrigation and water stress causes yield loss at any stage of the crop during the growing season (Çakir, 2004; Song et al., 2019). Therefore, meeting water demands by

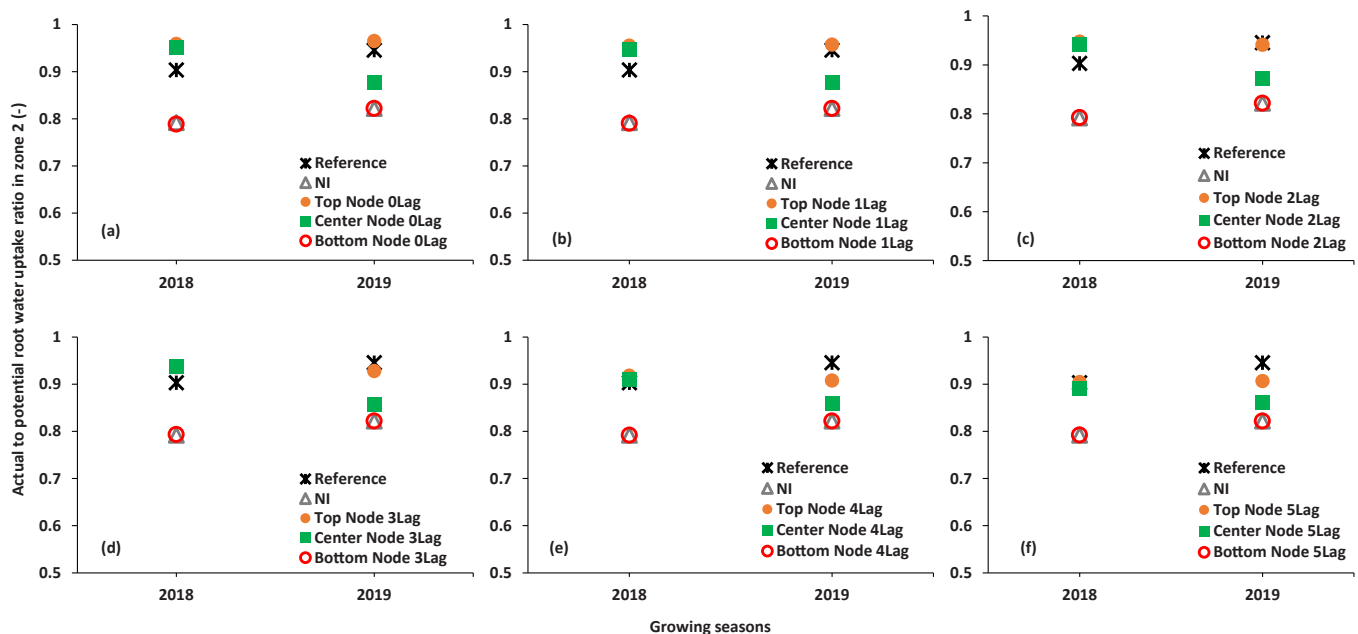


Fig. 8. Actual to potential root water uptake ratios in zone 2 for (a) 0Lag, (b) 1Lag, (c) 2Lag, (d) 3Lag, (e) 4Lag, and (f) 5Lag irrigation scenarios during 2018 and 2019 growing seasons. Reference scenario is the inverse model and NI is the model without irrigation data. Zone 2 is the delineated irrigation management zone within the field.

plants using zone-specific irrigation practices within a field can reduce the water stress and improve the crop yield during the growing season. Increasing the actual to potential root water uptake ratio can potentially result in increasing crop yield (Ramos et al., 2009, 2011).

The advantage of this proposed method is that we can optimize the irrigation thresholds using the modeling approach and data collected in each zone. Since irrigation thresholds are critical in sensor-based irrigation and depend on the accurate determination of SHPs, it is important to consider the soil water dynamics within the zones rather than using raw SHPs. The disadvantage of this method is that we need the field measurements of h or θ to optimize the SHPs and to determining the effective irrigation thresholds. The results obtained in this study can be helpful with the implication to the crop yield models (DSSAT) with different crop yield scenarios. Also, the determined irrigation recommendations can be used to trigger irrigation and quantifying crop yield. The proposed method of optimizing irrigation thresholds can be used for site-specific irrigations in the fields and in the modeling studies. It is important to know the correct upper limit of water holding capacity, i.e., field capacity in irrigation management, especially for site-specific irrigations. This static field capacity can be a source of error in this study since we assumed a field capacity of 33 kPa in both zones due to fine-textured soils. This study best represented the site-specific irrigation scenarios. Therefore, additional research is needed to validate the suggested irrigation scenarios incorporating crop yield and implications of the irrigation management scenarios investigated in this study. Also, the impact of zone-specific measurement of field capacity can be studied to improve the model and irrigation thresholds in future research.

4. Conclusions

The zones had spatial variability in SHPs along with the differences in soil texture, topographic attributes, and crop yield. The optimized inverse HYDRUS-1D model showed the impact on the shape and parameters of SHPs in both zones. Determination of irrigation thresholds was highly dependent on zone-specific characteristics of SHPs and important for site-specific management of irrigation water. The determined irrigation thresholds can be used for efficient VRI and other irrigation water management. Based on our simulations in this study using optimized models, the irrigation thresholds of around 80 kPa in zone 1 and 60 kPa in zone 2 can arrive at different times or at the same time during the growing season. The results showed conservation of irrigation water when using optimized SHPs compared with laboratory drawn raw SHPs in both zones to trigger irrigation. Due to the lack of water available to the plants, especially in the middle of the growing season, the crop can suffer from severe water stress. Farmers can improve site-specific irrigation practices in both zones considering spatial variability in SHPs, and 15 cm or 30 cm depths in zone 1 and 15 cm in zone 2 can be used as optimal depths to monitor the soil water pressure to trigger the irrigation. Improving the ARWU/PRWU ratio by scheduling irrigation when needed can increase the crop yield by meeting the crop water demands on time. Uniform irrigation or assuming one threshold for irrigation and the use of raw SHPs within the zones can be flawed irrigation strategies. Optimizing zone-specific SHPs showed that determining an accurate irrigation threshold in each zone can be effective irrigation practice.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgments

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.agwat.2022.107877](https://doi.org/10.1016/j.agwat.2022.107877).

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